Mathematics for ML: Numbers, Equations & Exponentials

Yogesh Haribhau Kulkarni

Outline

About Me

Yogesh Haribhau Kulkarni

Bio:

- 20+ years in CAD/Engineering software development
- Got Bachelors, Masters and Doctoral degrees in Mechanical Engineering (specialization: Geometric Modeling Algorithms).
- Currently doing Coaching in fields such as Data Science, Artificial Intelligence Machine-Deep Learning (ML/DL) and Natural Language Processing (NLP).
- Feel free to follow me at:
  - Github (github.com/yogeshhk)
  - LinkedIn (www.linkedin.com/in/yogeshkulkarni/)
  - Medium (yogeshharibhaukulkarni.medium.com)
  - Send email to yogeshkulkarni at yahoo dot com

Office Hours: Saturdays, 2 to 5pm (IST); Free-Open to all; email for appointment.

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About Content

- Bare essential mathematics needed for Machine/Deep Learning
- Just to get you started.
- Each field within Machine/Deep Learning can go extremely deep.
- This is to give you enough foundation needed.

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About Me

- I am not a mathematician.
- And this course won't turn YOU into one!!! (if you are not already)
- But, I will surely attempt to make you get interested in the learning, more.

Land of Mathematics

High School Mathematics

- Which Mathematics related words do you remember?
- Which Mathematics related branches do you know?
- Mathematics is the Language of Science.
- It provides grammar, words, structure.
- Our first encounter: Equations ...

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Numbers

Numbers

- $x = 3 \Rightarrow \mathbb{N}$: Natural numbers: 1,2,3,...
- $x + 5 = 3 \Rightarrow \mathbb{Z}$: Integers: -1,0,1,2,3,...
- $2x = 3 \Rightarrow \mathbb{Q}$: Rational numbers, ratios: $\frac{3}{2}, \frac{1}{3}, \ldots$
- $x^2 = 2 \Rightarrow \mathbb{P}$: Irrational numbers, cannot be expressed as ratios: $\sqrt{2}, \sqrt[3]{2}, \ldots$.
- $\pi, e \Rightarrow \mathbb{R}$: Real numbers, these cannot be represented by polynomials, cannot be roots, etc.
- $x^2 + 1 = 0 \Rightarrow \mathbb{C}$: Complex numbers: $i, 2 + 3i, \ldots$.All polynomials have roots in $\mathbb{C}$

$\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R} \subseteq \mathbb{C}$, with $\mathbb{P} = \mathbb{R} \setminus \mathbb{Q}$ (irrationals are the reals that are *not* rational – disjoint from $\mathbb{Q}$, not a superset of it)

Note: all the coefficients in the equation are $\mathbb{N}$ but the resultant x is of different types.

Intuition

Think of each new number system as "unlocking" a new kind of equation. Whole counting numbers can't solve $x + 5 = 3$ (needs negatives), integers can't solve $2x = 3$ (needs fractions), and rationals can't solve $x^2 = 2$ (needs irrationals). Each extension exists because some equation demanded it.

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Natural Numbers

- Natural numbers are the "natural" objects to count things around us with. The first thing we learn is to add natural numbers, then later on we start to multiply.
- Axioms for $\mathbb{N}$:

  Axiom N1 $1 \in \mathbb{N}$.
  Axiom N2 If $n \in \mathbb{N}$, then $n + 1 \in \mathbb{N}$.
  Axiom N3 There is no natural number $n \in \mathbb{N}$, such that $n + 1 = 1$.
  Axiom N4 For every natural number $n \in \mathbb{N}$, there is a real number $r \in \mathbb{R}$ such that $n \leq r < n + 1$.

Intuition

These axioms just formalize what you already know: 1 is a natural number, adding 1 to a natural number gives another one (so they never "run out"), you can't count backwards past 1, and every natural number sits between two consecutive reals. It's the rulebook, not new information.

(Ref: Helmut Knaust Class 200510 3341)

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Integers, Rational and Irrational Numbers

- Deficiencies of the system of natural numbers start to appear when we want to divide, the quotient of two natural numbers is not necessarily a natural number, or when we want to subtract, the difference of two natural numbers is not necessarily a natural number.
- This leads quite naturally to two extensions of the concept of number.
- The set of INTEGERS, denoted by $\mathbb{Z}$, is the set

$$\mathbb{Z} = \{0, 1, -1, 2, -2, 3, -3, \ldots\}.$$

The set of RATIONAL NUMBERS $\mathbb{Q}$ is defined as

$$\mathbb{Q} = \left\{ \frac{p}{q} \mid p, q \in \mathbb{Z} \text{ and } q \neq 0 \right\}.$$

- Real numbers that are not rational are called IRRATIONAL NUMBERS.

(Ref: Helmut Knaust Class 200510 3341)

Integers, Rational and Irrational Numbers

- The existence of irrational numbers, first discovered by the Pythagoreans in about 520 B.C., must have come as a major surprise to Greek Mathematicians:
- The square root of 2 is irrational. ($\sqrt{2}$ is the positive number whose square is 2.)

(Ref: Helmut Knaust Class 200510 3341)

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Operations

Addition and multiplication of rational and real numbers interact in a reasonable manner—the following DISTRIBUTIVE LAW holds:
For all $a, b, c \in \mathbb{R}$

$$(a + b) \cdot c = (a \cdot c) + (b \cdot c)$$

(Ref: Helmut Knaust Class 200510 3341)

Quick Check: Numbers

Think About It

Is 0 a natural number, an integer, both, or neither? And is every integer also a rational number – why or why not?

Answer

Conventions differ on whether $0 \in \mathbb{N}$, but 0 is always an integer ($\mathbb{Z}$). Every integer n is also rational, since it can be written as $n/1$ – so $\mathbb{Z} \subseteq \mathbb{Q}$.

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Equations

Algebra

- Equation 'equates' two expressions. E.g. $3x + 4 = 10$
- We solve equations for unknowns such as x
- x in the equation above have coefficient 3.
- Others like 4 and 10 are constants.

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Equations

$3x + 4 = 10$

- To Solve, we isolate x on one side by operations.
- $3x + 4 - 4 = 10 - 4$
- Although a short hand way is just to transfer 4 from one side to other and changing the sign.
- $3x = 6$
- In case of coefficients, you divide after crossing the $=$
- $3x/3 = 6/3$
- $x = 2$
- To test, plug $x = 2$, back in the original equation and see if both sides 'equate'.

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The Distributive Property

- $3(x+2) = 18$ is same as $3x + 6 = 18$
- Intuitively we multiply everything from outside, as if its a coefficient.
- Now instead of just 3 I have another expression to multiply $(3x+3)(x+2) = 18$
- Same distributive property works?

Two Variables

- $2y - 4x = 2$
- How to solve?
- Can we solve?

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Two Variables

- $2y - 4x = 2$
- Isolate y first
- $2y = 2 + 4x$
- $y = 1 + 2x$
- Thats y's definition in terms of x.
- For any x, y can be calculated.
- x, y if plotted form a line, thus the original equation is called as Linear Equation.

Intuition

Solving for y turns the equation into a "recipe": plug in any x, and it tells you the matching y. Plot enough of these (x, y) pairs and they trace a straight line – that's why it's called a Linear Equation.

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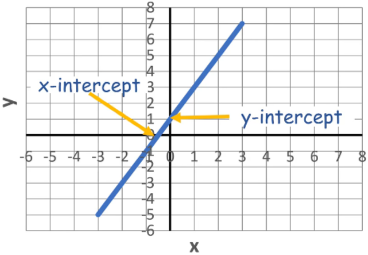
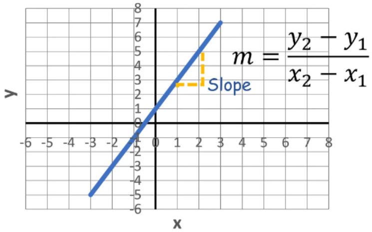
Line Plot

- X intercept is where line crosses x axis, where y is 0 (put in the eqn)
- Y intercept is where line crosses y axis, where x is 0 (put in the eqn)



Line Plot

- Slope is change in y per change in x
- Calculated using any two points.
- Slope-Intercept form is $y = mx + b$



Exercise

Find slope and both intercepts of $2y + 3 = 3x - 1$

Exercise

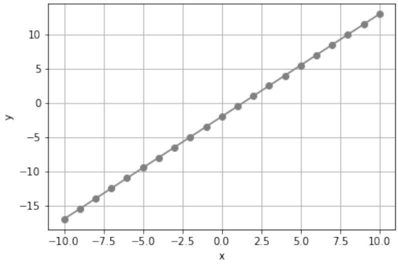
```python
import pandas as pd

# Create a dataframe with an x column containing values from -10 to 10
df = pd.DataFrame ({'x': range(-10, 11)})

# Add a y column by applying the solved equation to x
df['y'] = (3*df['x'] - 4) / 2

print(df)
```

Exercise

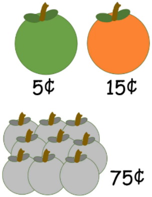



```python
from matplotlib import pyplot as plt

plt.plot(df.x, df.y, color="grey", marker = "o")
plt.xlabel('x')
plt.ylabel('y')
plt.grid()
plt.show()
```

System of Equations

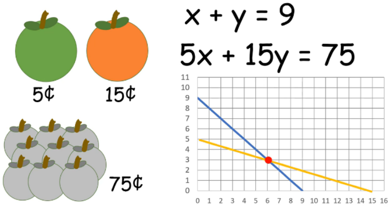
9 items cost 75c, how many are apples and how many oranges?



(Ref: Essentials of Mathematics - DAT 256 EdX)

System of Equations

Let number of apples be x and oranges be y.

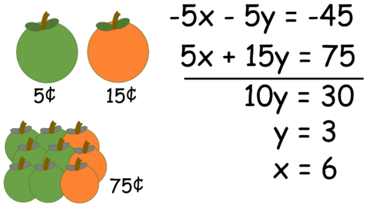


(Ref: Essentials of Mathematics - DAT 256 EdX)

Blue line represents $x + y = 9$ and yellow line $5x + 15y = 75$. Their intersection, the common point shared by two equations, is the solution.

System of Equations

System of Linear equations can be solved, apart from line intersection method, by elimination method.



(Ref: Essentials of Mathematics - DAT 256 EdX)

Make one of the equations (say, first) as NEGATIVE of the other in one variable, so that their addition cancels it out, leaving only one variable. Easy to find solution. Put that value back in the original equation, to get the other variable value.

Intuition

Elimination is just a shortcut for the graphical picture: two lines cross at exactly one point (usually). Instead of drawing both lines and reading off the crossing point, we algebraically cancel a variable so only one unknown remains

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Exercise

Solve $x + y = 16$ and $10x + 25y = 250$

Quick Check: Equations

Think About It

Two lines never intersect. What does that tell you about the number of solutions to their system of equations, and what does it mean geometrically about the lines?

Answer

Zero solutions – the system is inconsistent. Geometrically, the lines are parallel (same slope, different intercepts), so they never meet.

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Quick Check: Equations

Think About It

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Linear Equations

Linear Equations

Consider the simultaneous equations

$$\begin{aligned} x + 2y &= 4 \\ 3x - 5y &= 1 \end{aligned}$$

Provided you understand how matrices are multiplied together you will realize that these can be written in matrix form as

$$\begin{bmatrix} 1 & 2 \\ 3 & -5 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 4 \\ 1 \end{bmatrix}$$

Writing

$$A = \begin{bmatrix} 1 & 2 \\ 3 & -5 \end{bmatrix}, X = \begin{bmatrix} x \\ y \end{bmatrix}, \text{ and } B = \begin{bmatrix} 4 \\ 1 \end{bmatrix}$$

we have

$$AX = B$$

Intuition

This is the same "isolate x" idea from single-variable equations, just scaled up. Instead of dividing both sides by a number, we now "divide" by a matrix – which means multiplying by its inverse – to peel A away from X.

Linear Equations

We need to calculate the inverse of $A = \begin{bmatrix} 1 & 2 \\ 3 & -5 \end{bmatrix}$.

$$\begin{aligned} A^{-1} &= \frac{1}{(1)(-5)-(2)(3)} \begin{bmatrix} -5 & -2 \\ -3 & 1 \end{bmatrix} \\ &= -\frac{1}{11} \begin{bmatrix} -5 & -2 \\ -3 & 1 \end{bmatrix} \end{aligned}$$

Then X is given by

$$\begin{aligned} X = A^{-1}B &= -\frac{1}{11} \begin{bmatrix} -5 & -2 \\ -3 & 1 \end{bmatrix} \begin{bmatrix} 4 \\ 1 \end{bmatrix} \\ &= -\frac{1}{11} \begin{bmatrix} -22 \\ -11 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix} \end{aligned}$$

Hence $x = 2$, $y = 1$ is the solution of the simultaneous equations.

Linear Equations

m linear equations of n variables.

$a_{11}x_1 + a_{12}x_2 + \ldots + a_{1n} = u_1$

$a_{21}x_1 + a_{22}x_2 + \ldots + a_{2n} = u_2$

...

$a_{m1}x_1 + a_{m2}x_2 + \ldots + a_{mn} = u_m$

Can be written in matrix form as :

$$\begin{bmatrix} a_{11} & a_{12} & \ldots & a_{1n} \\ a_{21} & a_{22} & \ldots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \ldots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_m \end{bmatrix} = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_m \end{bmatrix}$$

Example

- x: price of pencil
- y: price of pen
- u: box 1
- v: box 2
- Set I:

$$2x + 3y = u \tag{1}$$
$$3x + 4y = v \tag{2}$$

- Set II:

$$2u + v = 57 \tag{3}$$
$$3u + 2v = 97 \tag{4}$$

Example

II by putting u, v from I

$$2(2x+3y)+1(3x+4y)=57 \tag{5}$$
$$3(2x+3y)+2(3x+4y)=97 \tag{6}$$

Collecting x, y

$$[(2x2)+(3x1)]x+[(2x3)+(4x1)]y=57 \tag{7}$$
$$[(2x2)+(3x1)]x+[(2x3)+(4x1)]y=97 \tag{8}$$

$$\begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} 2 & 3 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 57 \\ 97 \end{bmatrix}$$

Solving Simultaneous Equations

$$\begin{bmatrix} a_1 & a_2 & a_3 \\ a_4 & a_5 & a_6 \\ a_7 & a_8 & a_9 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix}$$

Make augmented matrix

$$\left[\begin{array}{ccc|c} a_1 & a_2 & a_3 & c_1 \\ a_4 & a_5 & a_6 & c_2 \\ a_7 & a_8 & a_9 & c_3 \end{array}\right]$$

Row operations permitted

- Interchange of rows
- Multiply by scalar
- Add rows

Using this make lower triable 0, and get the answers.

Assignment

- Q: Similar to row transformations can we apply column transformations as well? Difference?
- A: We can use column transformations as well but in that case the x vector changes. Eg. In case of $C_1 \iff C_3$, $\begin{bmatrix} x_3 \\ x_2 \\ x_1 \end{bmatrix}$ is the x vector.

A system of linear equations is a collection of equations in the same set of variables.
For example,

$$\begin{cases} x_1 + 3x_2 & = 5 \\ 2x_1 - x_2 & = -4 \end{cases}$$

- Pair of equations in two variables
- Think of this as the intersection of two lines.
- Sketch these lines and find their intersection.

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Solution Sets

The solution set for a linear system of equations can be:

- Empty. (there is no solution.)
- Exactly one solution
- More than one solution

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Solution Sets

- It is easy to come up with examples of the first two circumstances.
- The third possibility actually means that something much stronger is true.
- If we have two distinct solutions then we must have infinitely many solutions.
- Why is it that if we have at least two solutions then there are infinitely many?

To Do Draw two dimensional examples illustrating each of the three possible outcomes.

Matrix Notation

The system of equations above has coefficient matrix

$$\begin{bmatrix} 1 & 3 \\ 2 & -1 \end{bmatrix}$$

and augmented matrix

$$\begin{bmatrix} 1 & 3 & 5 \\ 2 & -1 & -4 \end{bmatrix}$$

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Elementary row operations

To Do Subtract twice the first row from the second row of the augmented matrix. Explain why the new system of equations has precisely the same set of solutions as the old set of equations.
Operations here are 3 types of elementary row operations.

- (Replacement) Replace a row by the sum of *the same row* and a multiple of *different* row.
- (Interchange) Interchange two rows.
- (Scaling) Multiply a row by a non-zero constant.

Notation

- (Replacement) The operation of replacing row i by itself plus c times row j will be denoted $R_i + cR_j$.
- (Interchange) The operation of interchanging rows i and j will be denoted by $R_i \leftrightarrow R_j$.
- (Scaling) The operation of scaling row i by a nonzero constant c will be denoted by cR_i.

It is clear that the set of solutions is not changed by the second and third operations.
To Do Explain why the set of solutions is unchanged by the first operation.

Equivalence of Linear Systems

Definition

- Two linear systems are **equivalent** if they have the same solution set.
- Two matrices are **row equivalent** if one matrix can be obtained from the other by elementary row operations.

Principle Suppose A and B are $m \times n$ matrices: explain why the following is true: if A can be reduced via elementary row operations to B, then B can be reduced to A.
The key fact to notice is that these operations can be undone or inverted.

Two linear systems are equivalent if and only if their augmented matrices are row-equivalent.
Write the following system of equations as an augmented matrix and solve the system:

$$\begin{array}{rcrcrcr} x_1 & + & & + & -3x_3 & = & 8 \\ 2x_1 & + & 2x_2 & + & 9x_3 & = & 7 \\ & + & x_2 & + & 5x_3 & = & -2 \end{array}$$

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Fundamental Questions

We can rephrase our question about how many solutions there are to a system in the following way.

- Is the system consistent, that is, do there exist *any* solutions?
- If there is at least one solution, is it unique?

Fundamental Questions

For example, we've seen that the system

$$\begin{aligned} x_1 + 3x_2 &= 5 \\ 2x_1 - x_2 &= -4 \end{aligned}$$

has a unique solution. On the other hand, the system

$$\begin{aligned} x_1 + 3x_2 &= 5 \\ 2x_1 + 6x_2 &= -4 \end{aligned}$$

has no solutions, and the system

$$\begin{aligned} x_1 + 3x_2 &= 5 \\ 2x_1 + 6x_2 &= 10 \end{aligned}$$

has infinitely many.

Row Reduction and Row Echelon Form

A rectangular matrix is in **echelon form** (or **row echelon form**) if it has the following properties

- All non-zero rows are above any rows of all zeros.
- Each leading entry of a row is in a column to the right of the leading entry of the row above it.
- All entries in a column below a leading entry are zeros.

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Row Reduction and Row Echelon Form

- Sometimes we want more than echelon form.
- We can make all the leading entries 1 by multiplying by a constant, and we can subtract from rows above to zero out their entries in that column.

Definition A rectangular matrix is in **reduced echelon form** (or **reduced row echelon form**) if it is in row echelon form, and has the following additional properties

- The leading non-zero term of every non-zero row is 1
- Each leading 1 is the only non-zero entry in its column.

Intuition

Echelon form is just "staircase" form – each row starts further right than the one above it, like a staircase going down. Reduced echelon form is the tidiest possible staircase: every step is exactly 1, and nothing else shares its column, so you can read off the answer directly.

Row Reduction and Row Echelon Form

Examples

$$\begin{bmatrix} 1 & 2 & 5 \\ 0 & 2 & 6 \\ 0 & 0 & 0 \end{bmatrix}$$

is in row echelon form but is not reduced:

$$\begin{bmatrix} 1 & 2 & 5 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix}$$

is in row echelon form but is not reduced:

Row Reduction and Row Echelon Form

Examples

$$\begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix}$$

is in reduced row echelon form:

$$\begin{bmatrix} 1 & 2 & 5 \\ 0 & 1 & 3 \\ 2 & 0 & 0 \end{bmatrix}$$

is not in row echelon form.

Row Reduction and Row Echelon Form

Solving a system in reduced row echelon form Suppose we have a system of equations, we've written them as an augmented matrix, we've performed elementary row operations, and arrived at the following reduced row echelon form matrix.

$$\begin{bmatrix} 1 & 6 & 0 & 3 & 0 & 0 \\ 0 & 0 & 1 & -4 & 0 & 5 \\ 0 & 0 & 0 & 0 & 1 & 7 \end{bmatrix}$$

- Write down the corresponding equations.
- Pair up the variables and the pivot columns: these are the **basic variables**.
- The remaining variables are the **free variables**.

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Great News

- Each matrix is row equivalent to one and only one reduced echelon matrix.
- Gaussian elimination: How do we get a matrix into row echelon or reduced row echelon form?

Quick Check: Linear Equations

Think About It

A system's augmented matrix row-reduces to a row of all zeros except the last entry, which is 6 (i.e. $0 = 6$). What does that tell you about the system's solutions?

Answer

The system is inconsistent – it has no solution. A row that reads $0x_1 + 0x_2 + \cdots = 6$ is a false statement, which means no values of the variables can satisfy every equation simultaneously.

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Gauss Elimination

Gaussian Elimination

- We start with the left-most non-zero column,
- Working to the right and from the top down.
- At each stage, we will be working with the portion of the matrix which is below or to the right (or both) of the pivot.

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Gaussian Elimination

- Find the left-most column containing a non-zero entry. This is a pivot column: the pivot position is at the top.
- Select a non-zero entry in the column to be the pivot. By interchanging rows if necessary, move the pivot into the pivot position.
- Use row replacement operations to change all the values below the pivot to zero.
- Move to the next row, and apply steps 1, 2, 3 to the remaining submatrix, namely the rows below and including the current row.

Once we've gone through all the rows, the matrix is in row echelon form.

Gaussian Elimination

- Scale pivots to be 1
- Use row replacement operations to change all the values above pivots to be zero. (There are technical reasons for doing this from the bottom pivot first).

Gaussian Elimination

In order to obtain row echelon form, we only need to switch rows and subtract multiples below.
This process is often referred to as Gaussian elimination, or Gauss-Jordan elimination

Example

Row reduce

$$\begin{bmatrix} 1 & 3 & 5 & 7 \\ 3 & 5 & 7 & 9 \\ 5 & 7 & 9 & 1 \end{bmatrix} \to \cdots$$

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Solutions of Linear Equations

Recall that there are three possible outcomes:

- No solutions: for example

$$\begin{bmatrix} 1 & 0 & 0 & 5 \\ 0 & 1 & 1 & 3 \\ 0 & 0 & 0 & 6 \end{bmatrix}$$

- Unique solution: for example

$$\begin{bmatrix} 1 & 0 & 0 & -5 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 6 \end{bmatrix}$$

- Infinitely many solutions: for example

$$\begin{bmatrix} 1 & 2 & 0 & 6 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

General Form

In the last case, we write the solution set as

$$\begin{cases} x_1 & = 6 - 2x_2 \\ x_2 & \text{is free} \\ x_3 & = 3 \end{cases}$$

- This is called a *general solution*.
- x_1 and x_3 are called dependent variables.
- x_2 is called a parameter or free variable or an independent variable.

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Find the general solution to the linear system with augmented matrix

$$\begin{bmatrix} 1 & 6 & 2 & -5 & -1 & -4 \\ 0 & 0 & 2 & -8 & -1 & 3 \\ 0 & 0 & 0 & 0 & 1 & 7 \end{bmatrix} \to \cdots$$

If a linear system is consistent then the solution set contains either

- A single element (-i.e. the solution is unique, there are no free variables).
- Infinitely many elements (-i.e. at least one free variable).

Solution Procedures Given a linear system to solve

- Write the augmented matrix
- Perform row reduction to obtain echelon form. If the system is not consistent then there are no solutions and you may stop.
- Perform row reduction to obtain reduced echelon form.
- Write system of equations corresponding to reduced echelon form.
- Basic variables correspond to columns with pivots. Free variables correspond to columns without pivots.
- Write basic variables in terms of free variables.

Definition If A is an $m \times n$ matrix with columns $\vec{a}_1, \vec{a}_2, \ldots, \vec{a}_n$, and $\vec{x} \in \mathbb{R}^n$, then the product of A and $\vec{x}$, which we denote by $A\vec{x}$, is defined to be

$$A\vec{x} = \begin{bmatrix} | & | & \cdots & | \\ \vec{a}_1 & \vec{a}_2 & \cdots & \vec{a}_n \\ | & | & \cdots & | \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

$$= x_1\vec{a}_1 + x_2\vec{a}_2 + \cdots + x_n\vec{a}_n$$

Note

- $A\vec{x} \in \mathbb{R}^m$.
- $A\vec{x}$ is the linear combination of the columns of A with weights $x_1, x_2, \ldots, x_n$.

Intuition

Don't think of $A\vec{x}$ as "rows times column". Think of it as a mixing recipe: take the columns of A as ingredients, and $x_1, x_2, \ldots$ as how much of each to mix in. The result is just a weighted blend of A's columns.

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Example

$$\begin{bmatrix} 1 & 2 \\ -1 & 3 \\ 3 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} =$$

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Theorem If A is an $m \times n$ matrix with columns $\vec{a}_1, \vec{a}_2, \ldots \vec{a}_n$, and if $\vec{b} \in \mathbb{R}^m$, then the matrix equation $A\vec{x} = \vec{b}$ has the same set of solutions as the vector equation

$$x_1\vec{a}_1 + x_2\vec{a}_2 + \cdots + x_n\vec{a}_n = \vec{b}$$

which in turn has the same set of solutions as the linear system with augmented matrix

$$\begin{bmatrix} | & | & \ldots & | & | \\ \vec{a}_1 & \vec{a}_2 & \ldots & \vec{a}_n & \vec{b} \\ | & | & \ldots & | & | \end{bmatrix}$$

YHK

Theorem The equation $A\vec{x} = \vec{b}$ has a solution if and only if $\vec{b}$ is a linear combination of the columns of A.

YHK

Example For which vectors $\vec{b}$ is the equation $A\vec{x} = \vec{b}$ solvable, where

$$A = \begin{bmatrix} 1 & 3 & -2 \\ 7 & 2 & 3 \\ -2 & 13 & -13 \end{bmatrix}?$$

Theorem Let A be an $m \times n$ matrix. The following statements are equivalent:

- For each $\vec{b} \in \mathbb{R}^m$, the equation $A\vec{x} = \vec{b}$ has a solution.
- Each $\vec{b} \in \mathbb{R}^m$ is a linear combination of the columns of A.
- The columns of A span $\mathbb{R}^m$.
- In the row reduction process, A has a pivot position in every row

Note Part 4 refers to the coefficient matrix A, not the augmented matrix $[A \quad \vec{b}]$.

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Proof Statements 1, 2 and 3 are equivalent by definition.
So if we show that 1 and 4 are equivalent, we will be done.

Suppose that U is the reduced echelon form of A.
Then for some vector $\vec{d}$, we have

$$[A \quad \vec{b}] \sim \cdots \sim [U \quad \vec{d}]$$

If statement 4 is true, then since U has a pivot in every row, we clearly don't have a row of zeros in U with a non-zero element in $\vec{d}$.
Hence the equation is solvable no matter what $\vec{b}$ is.
Thus, if statement 4 is true, then 1 is true.

Conversely, if 4 is false, U has a zero row.
Let $\vec{d}$ be the vector with a 1 in that row, and zeros elsewhere.
Reversing the row reduction process we obtain a vector $\vec{b}$ for which $A\vec{x} = \vec{b}$ has no solution.
Hence if statement 4 is false, so is statement 1.

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Computing $A\vec{x}$

In the computation of

$$\begin{bmatrix} a_{11} & \dots & a_{1n} \\ \vdots & & \vdots \\ a_{i1} & \dots & a_{in} \\ \vdots & & \vdots \\ a_{m1} & \dots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ \vdots \\ b_i \\ \vdots \\ b_m \end{bmatrix}$$

we see that

$$b_i = \sum_{j=1}^{n} a_{ij} x_j = a_{i1}x_1 + a_{i2}x_2 + \dots a_{in}x_n.$$

YHK

$$\begin{bmatrix} 1 & 3 & 5 \\ 2 & 4 & -1 \\ -1 & 2 & 2 \end{bmatrix} \begin{bmatrix} 11 \\ 1 \\ 3 \end{bmatrix} = \begin{bmatrix} \quad \\ \quad \\ \quad \end{bmatrix}$$

Theorem If A is an $m \times n$ matrix and $\vec{u}, \vec{v}$ are vectors in $\mathbb{R}^n$ and $c \in \mathbb{R}$ is a scalar then

- $A(\vec{u} + \vec{v}) = A\vec{u} + A\vec{v}$
- $A(c\vec{u}) = c(A\vec{u})$

YHK

Definition A system of linear equations is called *homogeneous* if it can be written as $A\vec{x} = \vec{0}$.
Note $\vec{x} = \vec{0}$ is always solution to $A\vec{x} = \vec{0}$. It is called the *trivial solution*.

Facts The homogeneous equation $A\vec{x} = \vec{0}$ has a non-trivial solution if and only if it has free variables.

Proof ($\Rightarrow$): Suppose that $A\vec{x} = \vec{0}$ has a nontrivial solution Then we have at least two solutions (-i.e. The trivial one and at least one other). Thus we must have an infinite number of solutions which requires a free variable.
($\Leftarrow$): Now suppose that when we row reduce the augmented matrix $[A : \vec{0}]$ that we have a free variable. Since we already know that we have a solution (the trivial one), and that we have a free variable, we must have infinitely many solutions . Thus we have nontrivial solutions.

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Exercise

- Determine if the following homogeneous system has non-trivial solutions:

$$\left\{\begin{array}{rcrcrcl} 2x_1 & + & 3x_2 & + & x_3 & = & 0 \\ & & 5x_2 & - & x_3 & = & 0 \\ -x_1 & + & x_2 & - & x_3 & = & 0 \end{array}\right.$$

- Describe the solution set.

Note The solution set of $A\vec{x} = \vec{0}$ can always be written as $\text{Span}(\vec{v}_1, \ldots, \vec{v}_p)$ for some vectors $\vec{v}_1, \ldots, \vec{v}_p$.

Definition An equation of the form

$$\vec{x} = s_1\vec{v}_1 + s_2\vec{v}_2 + \cdots + s_k\vec{v}_k$$

is said to be in *vector parametric form*.

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Example Describe the solution sets of $A\vec{x} = \vec{0}$ and $A\vec{x} = \vec{b}$ where

$$A = \begin{bmatrix} 3 & 5 & -4 \\ -3 & -2 & 4 \\ 6 & 1 & -8 \end{bmatrix} \quad \text{and} \quad \vec{b} = \begin{bmatrix} 7 \\ -1 \\ -4 \end{bmatrix}$$

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Theorem Suppose that the equation $A\vec{x} = \vec{b}$ has a solution $\vec{p}$. Then all solutions to the equation have the form

$$\vec{w} = \vec{p} + \vec{v}_h$$

where $\vec{v}_h$ is a solution to the corresponding homogeneous equation $A\vec{x} = \vec{0}$.
Note That is, if $\vec{p}$ is any solution to $A\vec{x} = \vec{b}$, and the solution set of $A\vec{x} = \vec{0}$ is $\mathrm{Span}(\vec{v}_1, \ldots, \vec{v}_k)$, then the solution set of $A\vec{x} = \vec{b}$ is

$$\vec{p} + \mathrm{Span}(\vec{v}_1, \ldots, \vec{v}_k)$$

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Definition An indexed set of vectors $\{\vec{v}_1, \ldots, \vec{v}_p\}$ in $\mathbb{R}^n$ is **linearly independent** if the vector equation

$$x_1\vec{v}_1 + \cdots + x_p\vec{v}_p = \vec{0}$$

has only the trivial solution ($\vec{x} = \vec{0}$).
Otherwise if there exist $c_1, \ldots, c_p \in \mathbb{R}$ not all zero, so that

$$c_1\vec{v}_1 + \cdots + c_p\vec{v}_p = \vec{0}$$

then the set is *linearly dependent*.

Intuition

"Linearly independent" just means no vector in the set is redundant – none of them can be built out of the others. If one vector IS just a combination of the rest (linearly dependent), it's not adding any new direction, only new arithmetic.

Example Are the vectors $\vec{v}_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$, $\vec{v}_2 = \begin{bmatrix} -1 \\ 1 \\ 5 \end{bmatrix}$, $\vec{v}_3 = \begin{bmatrix} -1 \\ 7 \\ 21 \end{bmatrix}$, linearly independent? If not, find a dependence.

Note The columns of the matrix A are linearly independent if and only if the equation $A\vec{x} = \vec{0}$ has only the trivial solution.

Example Are the columns of $A = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 2 & 2 \\ 5 & 3 & 3 \end{bmatrix}$ linearly independent?

Note

- A set containing a single vector $\vec{v}$ is linearly independent if and only if $\vec{v} \neq 0$.
- A set containing two vectors is linearly independent if and only if neither vector is a multiple of the other.

Theorem An indexed set $S = \{\vec{v}_1, \ldots, \vec{v}_p\}$ is linearly dependent if and only if one of the vectors in S is a linear combination of the others. In fact, S is linearly dependent if and only if either $\vec{v}_1 = \vec{0}$, or there is a j so that $\vec{v}_j$ is a linear combination of $\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_{j-1}$.

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Example Let $\vec{u} = \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix}$, $\vec{v} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$. Describe $\mathrm{Span}(\vec{u}, \vec{v})$. For this particular $\vec{u}, \vec{v}$ we have $\vec{w} \in \mathrm{Span}(\vec{u}, \vec{v})$ if and only if $\{\vec{u}, \vec{v}, \vec{w}\}$ is linearly dependent. Explain.

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Theorem If a set contains more vectors than there are entries (that is, rows) in the vectors, then it is linearly dependent.

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Theorem If $\vec{0} \in S = \{\vec{v}_1, \ldots, \vec{v}_p\}$ then S is linearly dependent.

Testing for linear dependence

To test whether vectors $\vec{v}_1, \vec{v}_2, \ldots \vec{v}_k$ are linearly dependent, construct the matrix $A = [\vec{v}_1, \vec{v}_2, \ldots \vec{v}_k]$ having them as columns. Perform row reduction on the matrix.
If every column contains a pivot, then the only solution to $A\vec{x} = \vec{0}$ is $\vec{x} = \vec{0}$, and hence the vectors are linearly independent, since then the only solution to

$$x_1\vec{v}_1 + x_2\vec{v}_2 + \cdots + x_k\vec{v}_k = \vec{0}$$

is $x_1 = x_2 = \cdots = x_k = 0$.

Testing for linear dependence

On the other hand, if there is a column without a pivot, then there are infinitely many solutions to the equation $A\vec{x} = \vec{0}$ (since there is a free variable in the general solution to the equation): pick a non-zero solution: it will correspond to a non-trivial solution to

$$x_1\vec{v}_1 + x_2\vec{v}_2 + \cdots + x_k\vec{v}_k = \vec{0}$$

and hence the vectors are linearly dependent.

Quick Check: Gaussian Elimination

Think About It

You row-reduce a 3×3 coefficient matrix A and find that one column has no pivot. What does that tell you about whether the columns of A are linearly independent?

Answer

They are linearly dependent. A column without a pivot means $A\vec{x} = \vec{0}$ has a free variable, so it has a nontrivial solution – which is exactly the definition of linear dependence.

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Quick Check: Gaussian Elimination

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Exponentials

Power

- $3 \times 3 = 3^2 = 9$
- $3 \times 3 \times 3 = 3^3 = 27$
- $4^7 = 16384$ here, 4 is the base, and 7 is the power or exponent in this expression.
- The 'raised to' can be seen as whole number.
- Can it be a fraction?
- Can it be negative?

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Radicals (Roots)

- Sometimes you'll need to calculate one or other of the elements themselves.
- $?^2 = 9$
- This expression is asking, given a number (9) and an exponent (2), what's the base?
- In other words, which number multiplied by itself results in 9?
- This type of operation is referred to as calculating the root, and in this particular case it's the square root $\sqrt{9} = 3$
- $36^{1/2} = ?$
- $81^{1/3} = ?$

Radicals (Roots)

```python
import math

# Calculate square root of 25
x = math.sqrt(25)
print (x)

# Calculate cube root of 64
cr = round(64 ** (1. / 3))
print(cr)
```

Logarithms

- To determine the exponent for a given number and base.
- In other words, how many times do I need to multiply a base number by itself to get the given result.
- $4^x = 16$, what is x?
- Such solutions are found by logarithms
- $x = \log_4(16) = 2$
- Log finds the POWER to given base, of a number.
- Whats Log of 1000 to base 10?

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Logarithms

```python
import math

# Natural log of 29
print (math.log(29))

# Common log of 100
print(math.log10(100))
```

Logarithms

- We know of log to base 10 and those log tables. But is there a function/formula?
- What is a log? Is it defined for all real values?
- $10^{2.3} = 10^{\frac{23}{10}}$ Thats 10th root of 10^{23}
- $10^{-0.07} = \frac{1}{10^{0.07}}$
- $10^{\sqrt{2}} = 10^{(2^{1/2})}$. Need to approximate $\sqrt{2}$ to some decimal places and then evaluate.
- $10^x = y$ then $\log y = x$
- x can be any real number and y will be corresponding unique number.

Intuition

A logarithm just answers "how many times do I multiply the base by itself to get here?" It's the exact opposite of exponentiation – exponentiation grows a number fast, log measures how much it grew.

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Quick Check: Exponentials & Logarithms

Think About It

If $\log_2(x) = 10$, roughly how big is x – tens, hundreds, or over a thousand? What does this tell you about how logarithms compress large numbers?

Answer

$x = 2^{10} = 1024$, over a thousand. A modest-looking log value (10) corresponds to a much larger original number – that's why logarithms are used to compress huge ranges (like earthquake magnitudes or sound decibels) into manageable scales.

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Quick Check: Exponentials & Logarithms

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Thanks . . .

- Office Hours: Saturdays, 3 to 5 pm (IST); Free-Open to all; email for appointment to yogeshkulkarni at yahoo dot com
- Call + 9 1 9 8 9 0 2 5 1 4 0 6

(https://medium.com/@yogeshharibhaukulkarni)

(https://www.linkedin.com/in/yogeshkulkarni/)

(https://www.github.com/yogeshhk/)

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